

Solution to Homework 5

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1 Exercise 5.1 - - Gram Schmidt Process

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <math.h>
4 #include <mpi.h>
5 #include <string.h>
6
7 #define N 65536
8 #define M 10
9 #define FILE_PATH "/work/korzec/LAB2/ex5/x_%d.dat"
10
11 void read_vector(float *vec, int id) {
12     char filename[50];
13     sprintf(filename, FILE_PATH, id);
14     FILE *file = fopen(filename, "r");
15     fread(vec, sizeof(float), N, file);
16     fclose(file);
17 }
18
19 float dot_product(float *a, float *b, int n_bar) {
20     float sum = 0.0;
21     for (int i = 0; i < n_bar; i++) {
22         sum += a[i] * b[i];
23     }
24     return sum;
25 }
26
```

```

27 void normalize(float *a, int n_bar) {
28     float norm = sqrt(dot_product(a, a, n_bar));
29     for (int i = 0; i < n_bar; i++) {
30         a[i] /= norm;
31     }
32 }
33
34 int main(int argc, char **argv) {
35     int rank, size;
36     MPI_Init(&argc, &argv);
37     MPI_Comm_rank(MPLCOMM_WORLD, &rank);
38     MPI_Comm_size(MPLCOMM_WORLD, &size);
39
40     int n_bar = N / size;
41     float *x = (float *)malloc(n_bar * sizeof(float));
42     float *b = (float *)malloc(n_bar * sizeof(float));
43
44     for (int l = 0; l < M; l++) {
45         if (rank == 0) {
46             read_vector(x, l);
47         }
48         MPI_Bcast(x, n_bar, MPI_FLOAT, 0,
49                 MPLCOMM_WORLD);
50
51         for (int k = 0; k < l; k++) {
52             float dot = dot_product(b, x, n_bar);
53             MPI_Allreduce(MPI_IN_PLACE, &dot, 1,
54                 MPI_FLOAT, MPLSUM, MPLCOMM_WORLD);
55             for (int i = 0; i < n_bar; i++) {
56                 x[i] -= b[i] * dot;
57             }
58
59             normalize(x, n_bar);
60             memcpy(b, x, n_bar * sizeof(float));
61         }
62
63         free(x);
64         free(b);
65         MPI_Finalize();

```

```
65     return 0;  
66 }
```

For compilation on Stromboli, we used the following command:

```
mpicc -o ex5 ex5.c
```

2 Submit Script for exercise 5.1

```
1 #!/bin/bash
2 #SBATCH --nodes=3
3 #SBATCH --ntasks=64
4 #SBATCH --partition=compute2011
5 #SBATCH --exclusive
6
7 module load mpi/openmpi/4.1.0
8 mpirun ./ex5
```

3 Results

```
[node13:787800:0:787800] Caught signal 11 (Segmentation fault: address not mapped to
malloc(): corrupted top size
[node13:787800] *** Process received signal ***
[node13:787800] Signal: Aborted (6)
[node13:787800] Signal code: (-6)
slurmstepd: error: *** JOB 105196 ON node13 CANCELLED AT 2023-11-11T21:23:50 ***
```